Public Health England

Principles of Outbreak Management

Soili Larkin & Joshna Mavji

Aim

Understand the principles of outbreak management

Objectives

- Define an outbreak
- Describe the purpose of outbreak management
- Outline the steps involved in the management of an outbreak
- Outline the roles and responsibilities of an Outbreak Control Team in the management of an outbreak
- Apply the learning to an outbreak exercise

What is an Outbreak?

What is an Outbreak?

- Two or more linked cases of an illness
- A single case of an unusual or rare infection.
- A greater than expected rate of infection compared with the usual background rate that is expected in that population for that place or time.
- A suspected, anticipated or actual event involving microbial contamination of food or water

Purpose of Outbreak Management

- Identify and control outbreaks of infection
- Reduce illness and complications
- Inform future prevention strategies
- Evaluate existing prevention strategies
- Address public concern

Outbreak Management Framework

- Case definition
- Confirm diagnosis in known cases
- Determine background incidence of disease
- Case-finding
- Data collection
- Descriptive epidemiology
- Microbiological investigation
- Environmental investigation
- Hypothesis generation
- Analytic epidemiology
- Communications
- Control measures

Outbreak Management: Case Definition

Case definition

Confirm diagnosis

Background incidence

Case-finding

Data collection

Descriptive epidemiology

Microbiological investigation

Environmental investigation

Hypothesis generation

Analytic epidemiology

Communications

Control Measures

Not always possible to secure laboratory confirmation:

- False negatives and positives not uncommon
- Samples are not always taken (e.g. diarrhoea)
- Appropriate tests may be unavailable (e.g. botulism)

Conversely, laboratory confirmation sometimes essential:

- e.g. legionella, hepatitis A/B/C, VHF
- Possible, probable, confirmed

Components

- Time…..based on incubation (if known)
- Place….location of suspected exposure
- Person….type of illness

Task – Using the components listed above, provide a case definition for an outbreak of Hepatitis A

Outbreak Management: Confirm Diagnosis

Case definition

Confirm diagnosis

Background incidence

Case-finding

Data collection

Descriptive epidemiology

Microbiological investigation

Environmental investigation

Hypothesis generation

Analytic epidemiology

Communications

Control Measures

- To assist in verifying whether an outbreak has occurred
- For describing the epidemiology
- Will involve verifying that cases fit the case definition
- Use medical records or laboratory reports

Outbreak Management: Background Incidence

Case definition

Confirm diagnosis

Background incidence

Case-finding

Data collection

Descriptive epidemiology

Microbiological investigation

Environmental investigation

Hypothesis generation

Analytic epidemiology

Communications

Control Measures

- Some outbreaks defined when rates exceed the background incidence in the community (e.g. cryptosporidiosis)
- Obtained from local and national surveillance data

Outbreak Management: Case Finding

Case definition

Confirm diagnosis

Background incidence

Case finding

Data collection

Descriptive epidemiology

Microbiological investigation

Environmental investigation

Hypothesis generation

Analytic epidemiology

Communications

Control Measures

Initial cases may only make up a small proportion of all cases. Active case finding helps:

- Establish the onset (which helps identify source)
- Define the extent of the outbreak – minor or major
- Identify cases requiring healthcare input
- Identify where control measures are needed
- Provide subjects for epidemiological studies
- Monitor effectiveness of control measures
- Determine when the outbreak can be declared over

Line listing is useful to summarise the information. It usually includes:

- Demographic information
- Clinical information
- Exposure information

Outbreak Management: Case Finding

Case definition

Confirm diagnosis

Background incidence

Case finding

Data collection

Descriptive epidemiology

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Communications

Control Measures

Additional cases can be sought from:

- Statutory notifications
- Microbiology laboratory results
- GPs
- Hospitals (A&E, ID Dept, inpatients, outpatients)
- Occupational health - large employers
- School reports of absenteeism and illness
- Household enquiries
- Media appeals
- Screening of population subgroups

Outbreak Management: Data Collection

Case definition

Confirm diagnosis

Background incidence

Case finding

Data collection

Descriptive epidemiology

Microbiological investigation

Environmental investigation

Hypothesis generation

Analytic epidemiology

Communications

Control Measures

- Data collection should be standardised
- Methods depend on the outbreak and resources available:
  - Detailed questionnaires: face-to-face administration
  - Mailed questionnaires for self-completion
  - Telephone interviews

Outbreak Management: Data Collection

Case definition

Confirm diagnosis

Background incidence

Case finding

Data collection

Descriptive epidemiology

Microbiological investigation

Environmental investigation

Hypothesis generation

Analytic epidemiology

Communications

Control Measures

- Data could include, as appropriate, questions about the following:
  - Name, age, sex;
  - GP details;
  - Occupation; work or school address;
  - Illness onset and clinical description;
  - Immunisation history;
  - Food history;
  - Water exposure;
  - Social activity: gardening, social events, social venues;
  - Sexual contacts;
  - Injecting drug use;
  - Close contacts: household, school, workplace;
  - Animal contact: pets, farm visits, travel;
  - Foreign or local travel: vaccines and drug history; departure & return dates, countries visited; places visited; accommodation; source of food and water; activities (work, leisure, trekking, jungle excursions, etc.); contact with animals.

Outbreak Management: Descriptive Epidemiology

Case definition

Confirm diagnosis

Background incidence

Case finding

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Microbiological investigation

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Communications

Control Measures

- Defines the outbreak in terms of **person**, **place** and **time**
- Provides clues re:- causative organism
- Hypothesis generation – source of infection / mode of transmission
- Define extent of the outbreak
- Monitor outbreak progress & control measures
- Helps decide when the outbreak can be declared over
- Epi Curves are useful to suggest:
  - Type of exposure
  - Time of exposure
  - Possible exposure agent

Outbreak Management: Descriptive Epidemiology

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Confirm diagnosis

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Control Measures

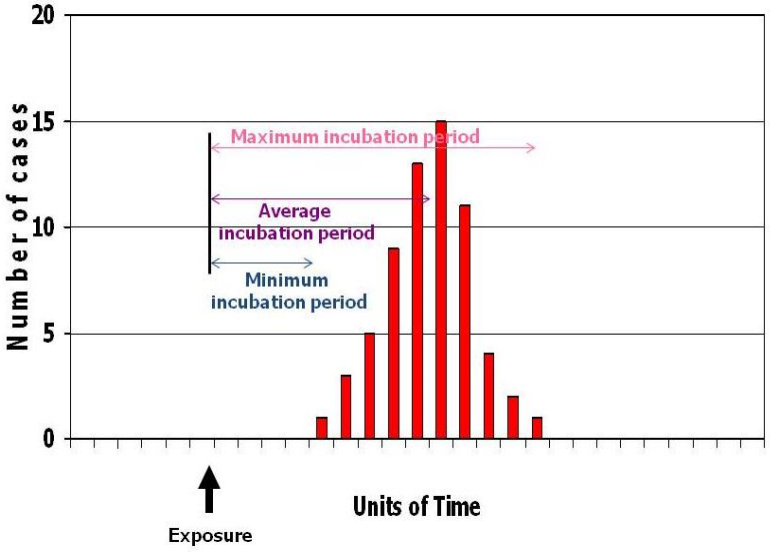
Patterns of Spread

- Common Source Outbreaks - Cases of disease arise from a single, shared or 'common' source, such as a batch of bad food, industrial pollution or a contaminated water supply. Controlling the source stops the outbreak.
  - **Point source outbreak** – all cases appear to occur within one incubation period therefore is usually of short duration
  - **Continuing source outbreak** – all cases are exposed to a single noxious influence but the exposure continues over a longer time so the outbreak persists for longer
  - **Intermittent outbreak** – all cases are exposed to a common source that is not well controlled, so outbreaks recur.
- Person-to-Person Spread – Disease spreads via person-to-person contact. Controlling the source is no longer sufficient to control the outbreak
  - **Index case with limited spread / point source with secondary transmission** – a single index case infects other individuals and cases arise after an incubation period. The outbreak wanes when the infected people no longer transmit infection to other susceptible individuals, usually because of implementation of successful control measures.
  - **Propagated spread** – index case infects other individuals (secondary cases) who, in turn, infect more individuals.

Outbreak Management: Descriptive Epidemiology

TASK

Match the epidemic curves with the type of outbreak they represent and give an example of each type



<u>Type of Outbreak</u>

Point Source outbreak

Continuing Source outbreak

Intermittent outbreak

Point Source with Secondary Transmission

Propagated Spread

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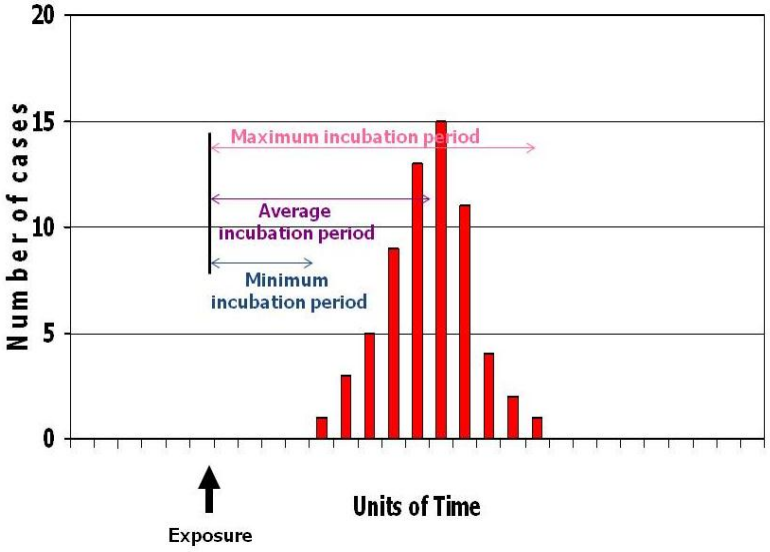

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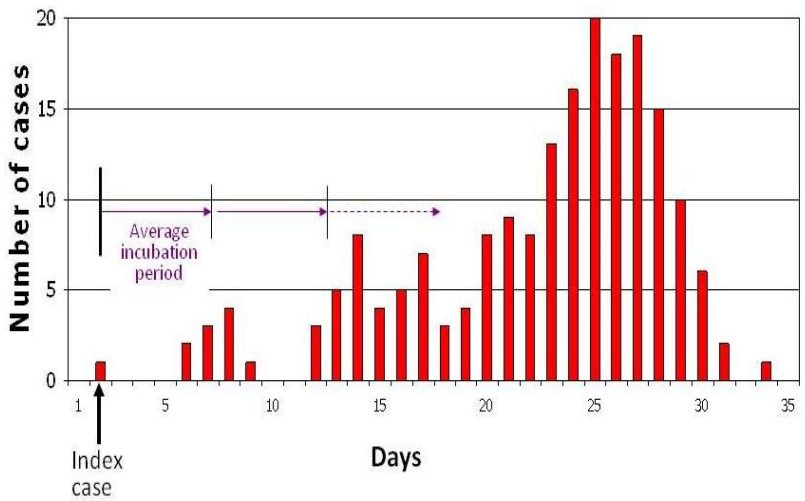

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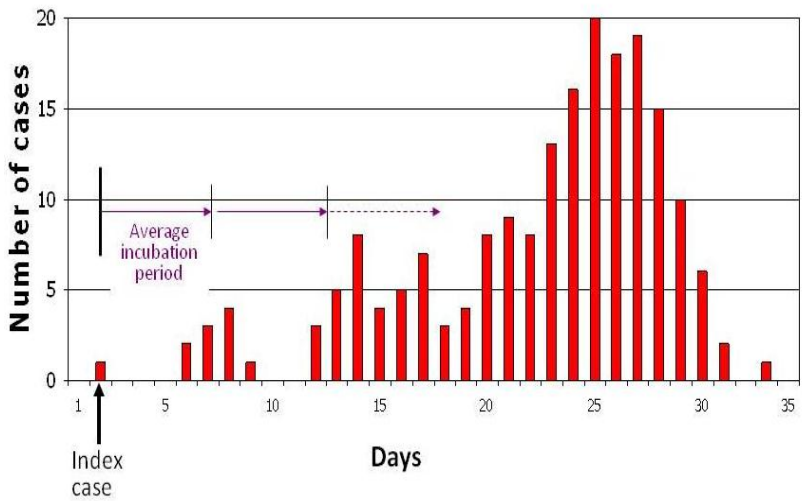

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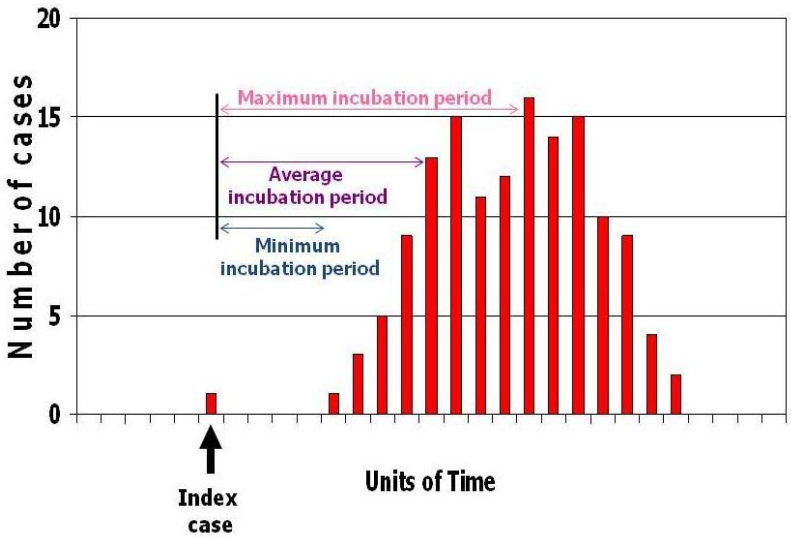

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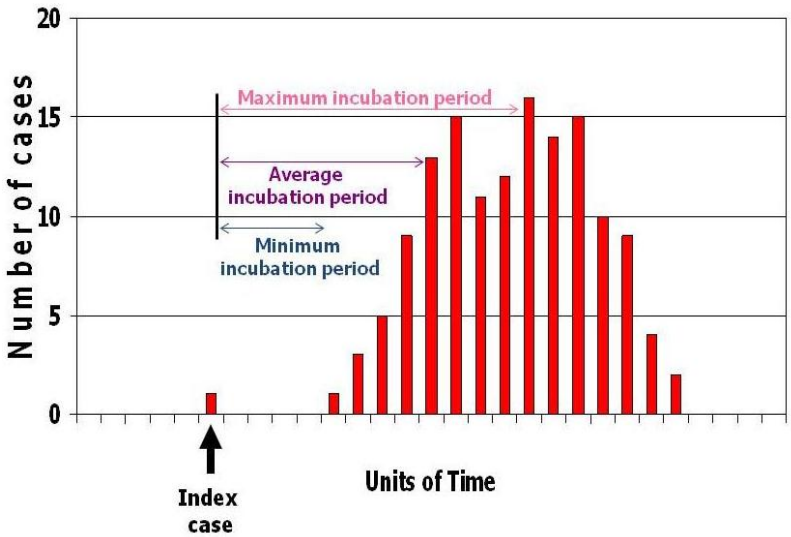

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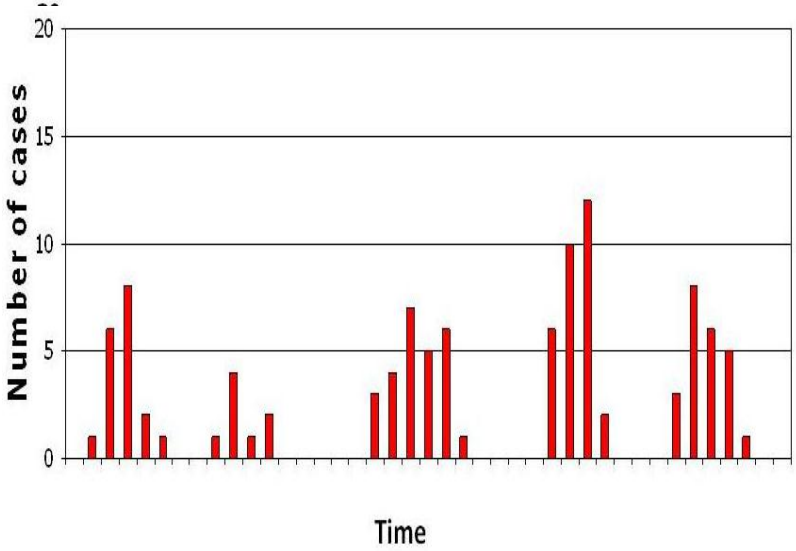

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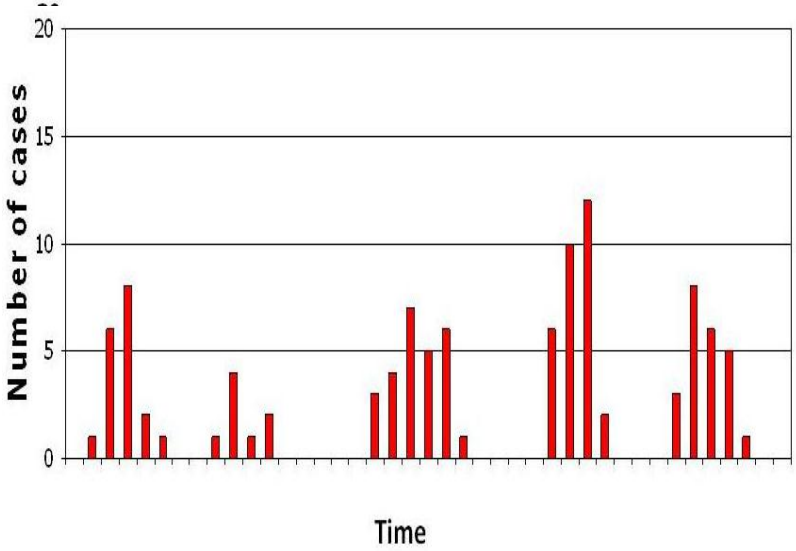

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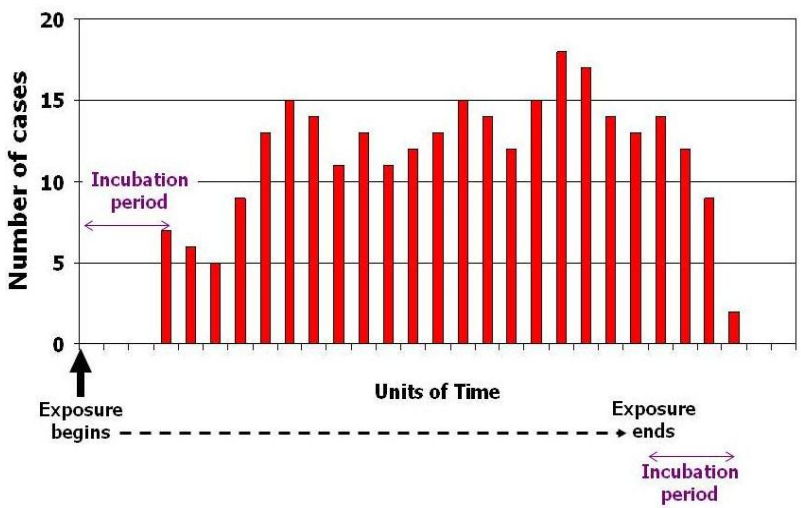

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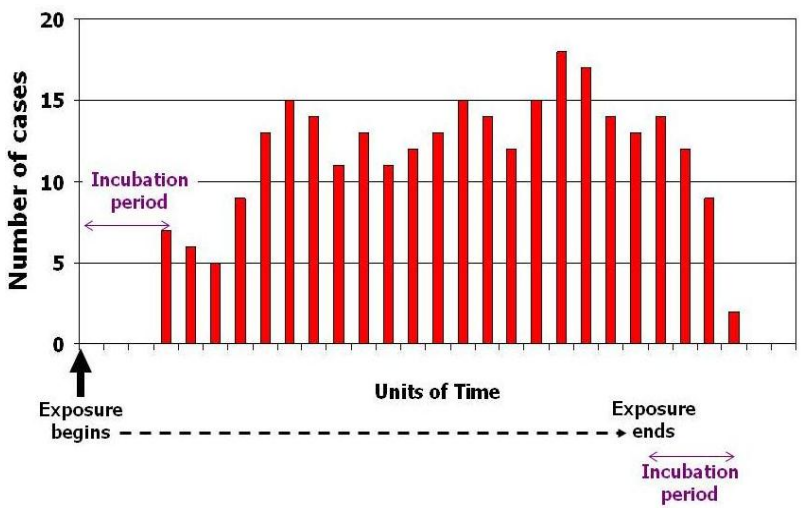

Type of Outbreak

Point Source outbreak

Continuing Source outbreak

Intermittent outbreak

Point Source with Secondary Transmission

Propagated Spread

Outbreak Management: Microbiological Investigation

Case definition

Confirm diagnosis

Background incidence

Case finding

Data collection

Descriptive epidemiology

Microbiological investigation

Environmental investigation

Hypothesis generation

Analytic epidemiology

Communications

Control Measures

- Laboratory confirmation not always possible
- Liaison with microbiologists (and veterinarians) to discuss:
    - Implications of laboratory results
    - Further human, (animal) and environmental samples

Outbreak Management: Environmental Investigation

Case definition

Confirm diagnosis

Background incidence

Case finding

Data collection

Descriptive epidemiology

Microbiological investigation

Environmental investigation

Hypothesis generation

Analytic epidemiology

Communications

Control Measures

Includes:

- Inspection/ sampling of physical environment (e.g. cooling towers)
- Appraisal of procedures and activities (e.g. personal hygiene, food hygiene and infection control policy)

Outbreak Management: Hypothesis Generation

Case definition

Confirm diagnosis

Background incidence

Case finding

Data collection

Descriptive epidemiology

Microbiological investigation

Environmental investigation

Hypothesis generation

Analytic epidemiology

Communications

Control Measures

- Need to consider epidemiological, microbiological and environmental evidence
- May be revised repeatedly as more evidence becomes available
- Credibility of hypothesis evaluated by:
  - Comparing with established facts
  - Analytic epidemiology - unnecessary if evidence strongly supports hypothesis

Outbreak Management: Analytic Epidemiology

Case definition

Confirm diagnosis

Background incidence

Case finding

Data collection

Descriptive epidemiology

Microbiological investigation

Environmental investigation

Hypothesis generation

Analytic epidemiology

Communications

Control Measures

- If resources available
- To further test hypotheses when doubts persist or control measures fail
- Comparison group: quantify exposure-disease relationship
- Statistical evaluation of attack rates or odds ratios
- Cohort studies include EVERYONE who could have been exposed therefore requires a complete list e.g. meeting/wedding attendees etc. Measure of association = relative risk
- Case-control studies compare exposures among ill persons (cases) and non-ill persons (controls). This is used when a complete list is not available or is too large e.g. restaurant attendees, national outbreaks etc. Measure of association = Odds Ratio

Outbreak Management: Communications

Case definition

Confirm diagnosis

Background incidence

Case finding

Data collection

Descriptive epidemiology

Microbiological investigation

Environmental investigation

Hypothesis generation

Analytic epidemiology

Communications

Control Measures

Robust communication systems are required for:

- Investigation – active case finding (e.g. from GPs, microbiologists)
- Control – disseminating advice to the public, health professionals and other agencies
- Liaison – with public, media, and health & non-health agencies

The Telegraph

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Germany's E.coli outbreak in pictures

First they pointed a finger at Spanish cucumbers. Then they cast suspicion on sprouts from Germany. Now German officials appear dumbfounded as to the source of the deadliest E. coli outbreak in modern history. Backtracking for the second time in a week, officials said that preliminary tests have found no evidence that vegetable sprouts from an organic farm in northern Germany were to blame. The surprise U-turn came only a day after the same state agency held a news conference to announce that the sprouts appeared to be the culprit in the outbreak that has killed 22 people and made more than 2,330 others sick across Europe, most of them in Germany, over the past month.

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15 June 2012 Last updated at 00:20

Share

Edinburgh Legionnaires' outbreak: Second victim dies

A second person has died after contracting Legionnaires' disease in Edinburgh.

The man, who was in his 40s, was from the Gorgie area of the city, which is at the centre of the outbreak.

It is understood the man had underlying health issues and was being treated at Edinburgh Royal Infirmary.

The total number of confirmed cases of Legionnaires' disease currently stands at 41.

SPL

It is thought the infection was spread from contaminated vapour from a cooling tower

Outbreak Management: Communications

Case definition

Confirm diagnosis

Background incidence

Case finding

Data collection

Descriptive epidemiology

Microbiological investigation

Environmental investigation

Hypothesis generation

Analytic epidemiology

Communications

Control Measures

Consideration of content:

- Appropriate for target audience
- Accurate
- Consistent
- Understandable (consider non-English speakers, blind, deaf etc.)

Methods include:

- Media (print, radio, TV)
- Telephone advice lines (in liaison with NHS Direct)
- Loudspeaker vans
- Leaflets
- Internet

Outbreak Management: Control Measures

Case definition

Confirm diagnosis

Background incidence

Case finding

Data collection

Descriptive epidemiology

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Hypothesis generation

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Communications

Control Measures

Basic principles:

- Removing source of infection
  - Isolate cases
  - Product recall
- Interrupting transmission
  - Personal hygiene e.g. hand washing
- Protecting persons at risk
  - Prophylaxis
  - Vaccination
- Preventing recurrence
  - Guidelines
  - Recommendations

Outbreak Management: Control Measures

Case definition

Confirm diagnosis

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Communications

Control Measures

TASK

What control measures could be put into place for the following:

- **Food poisoning outbreak**
- **Pandemic flu**

Basic Principles

- Removing the source of infection
- Interrupting transmission
- Protecting persons at risk
- Preventing recurrence

Public Health England

OUTBREAK CONTROL TEAM (OCT)

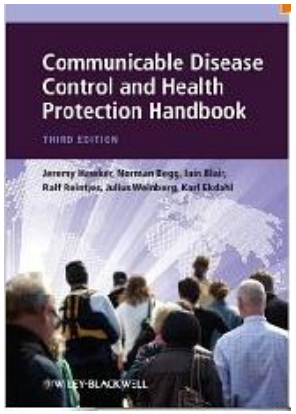
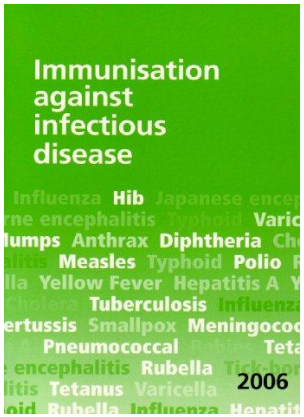
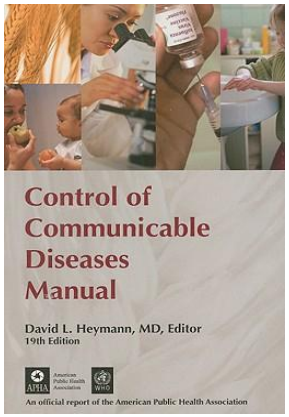
Outbreak Control Team: Roles and Responsibilities

- Review outbreak evidence – results of epidemiological, microbiological and environmental investigations
- Recommend control measures based on risk assessment
- Agree further investigations
- Decide on incident room establishment
- Decide on OCT membership
- Assign individual responsibilities to OCT members
- Determine resources required
- Ensure surveillance data to monitor progress
- Decide who is to be kept informed of the progress of the outbreak and circulate regular reports as appropriate
- Ensure arrangements to communicate with public/media
- Meet regularly during the outbreak and ensure that a written record of each meeting is made
- Decide criteria for declaring the outbreak over
- Produce and circulate a final report

Outbreak Control Team: Membership

- Chair (usually a CCDC)
- Environmental Health
- Public Health Laboratory representative
- Consultant Microbiologist
- Administrative and secretarial support
- Director of Public Health (or nominated deputy)
- Consultant Epidemiologist
- Communications Officer
- Health Protection Nurse/Practitioner
- Community Infection Prevention & Control
- Hospital Infection Prevention & Control
- Clinical Virologist
- Toxicologist
- CCG Representative
- NHS England Area Team Representative
- Representatives from affected NHS Trust
- Health & Safety Executive representative
- Food Standards Agency representative
- Environment Agency representative
- Department for Environment, Food & Rural Affairs representative
- State Veterinary Service representative
- Water Company representative
- Legal Officer

Public Health England

Further Reading

- Hawker, Begg *et al.,* (2012) *Communicable Disease Control and Health Protection Handbook*. 3rd edition.
- Public Health England. *Immunisation against infectious disease*. (online) https://www.gov.uk/government/collections/immunisation-against-infectious-disease-the-green-book
- Heymann. (2008) *Control of Communicable Diseases Manual*. 19th edition.